Data Flow

Food recognition

1. User uploads an image and optionally records voice or types text.
2. Frontend sends the files to the backend.
3. Backend validates type (JPG, PNG, WEBP) and size.
4. Image is resized to 256, center-cropped to 224 and normalized.
5. The PyTorch model predicts the dish and confidence.
6. The nutrition lookup returns calories, macros and a health score for that dish.
7. Frontend shows dish, confidence, serving, calories range, macros, health score.

Meal planning

1. User fills in the profile form (age, gender, height, weight, activity, goal, diet, cuisine, allergies, medical conditions, recent food).
2. Backend builds a prompt from the inputs.
3. Prompt is sent to Ollama (Llama).
4. The generated plan is returned and shown in the right panel.
5. If Ollama fails, the UI shows the error message.

Freshness

1. User uploads a food image.
2. Backend runs the freshness model and returns a fresh or rotten result with confidence. Low-confidence cases can be reported as uncertain.

Hygiene

1. Frontend loads the hygiene checklist from the backend.
2. User ticks the items followed and submits them.
3. Backend returns a hygiene score.

Smart Kitchen

1. User adds kitchen items with a name and expiry date, lists them, or deletes them.
2. For recipes, user sends a dish name or ingredients and gets a recipe back.

Voice

1. User uploads an audio file.
2. Backend transcribes it and returns the text.